

QBITEL BRIDGE

Quantum-Safe Infrastructure Security

TELECOMMUNICATIONS & 5G NETWORKS

Carrier-Grade Post-Quantum Cryptographic Security

SS7/Diameter Protection | 5G Core Security | IoT at Scale

150,000+

PQC ops/sec

99.999%

Availability SLA

<1 sec

SS7 Block Speed

5G Slice

Ready

EXECUTIVE SUMMARY

Carrier-Grade Quantum-Safe Security for Global Telecommunications

The global telecommunications industry operates the most critical digital infrastructure on earth — carrying voice, data, financial transactions, emergency services, and government communications for billions of people. Yet the protocols underpinning this infrastructure were designed in an era when security was an afterthought, and the quantum threat is transforming yesterday's theoretical risks into today's operational emergencies.

150,000+

PQC ops/sec

99.999%

Availability

<1 sec

SS7 Block

5G Slice

Ready

QBITEL Bridge delivers carrier-grade post-quantum cryptographic (PQC) security purpose-built for telecommunications networks: protecting SS7/Diameter signaling, 5G core network slices, SIP/VoIP infrastructure, IoT mass-device gateways, and subscriber data at 150,000+ cryptographic operations per second with 99.999% availability — without requiring updates to a single subscriber device.

In 2024 alone, 850 million SS7 attacks were recorded globally. International Revenue Share Fraud (IRSF) costs the industry more than \$10 billion annually. And the quantum threat to subscriber databases — holding biometric, financial, and location data for billions of people — is no longer measured in decades but in years.

QBITEL Bridge integrates directly into existing signaling infrastructure, 5G core network functions, and fraud management systems — providing a unified cryptographic defense layer that scales from regional MVNOs to the largest Tier-1 MNOs on the planet.

THE TELECOM SECURITY PARADOX

Networks that connect the world are themselves vulnerable

Telecommunications carriers face a fundamental paradox: the very protocols that enabled global connectivity were engineered for reliability and interoperability — not security. SS7, the signaling protocol suite that routes calls and SMS for billions of subscribers worldwide, was designed in 1975 with zero authentication mechanisms. Any operator connected to the global SS7 network can potentially send any message on behalf of any other operator.

This is not a theoretical concern. SS7 attack kits are commercially available on dark web forums for as little as \$500 per session. A sophisticated adversary can track the location of a VIP subscriber, intercept SMS-based two-factor authentication codes, forward calls silently, or deny service to a target — all from a laptop, anywhere in the world, exploiting protocols your network uses every second.

Meanwhile, the rollout of 5G networks has tripled the attack surface compared to 4G. The 5G Service-Based Architecture (SBA) introduces HTTP/2-based interfaces between network functions — bringing web-application vulnerabilities into the core of carrier infrastructure. Network slicing enables unprecedented service differentiation, but each slice boundary is a potential lateral movement vector.

And looming over all of this is the quantum threat. Nation-state adversaries are harvesting encrypted subscriber data today using harvest-now, decrypt-later strategies — collecting authentication vectors, location histories, and communications metadata that will become readable the moment a cryptographically-relevant quantum computer comes online.

THREE CRITICAL THREATS FACING TELECOM OPERATORS

THREAT	VECTOR	IMPACT	QBITEL BRIDGE RESPONSE
SS7/Diameter Legacy	MAP/ISUP/TCAP exploits, no authentication, globally accessible	850M attacks/year, location tracking \$500/target, SMS MFA bypass	Real-time MAP filtering, PQC auth vector wrapping, <1 sec block
5G Hyperscale Surface	SBI HTTP/2 APIs, cross-slice lateral movement, PFCP manipulation	3x larger than 4G, core NF compromise exposes millions of subscribers	Per-slice PQC policies, SBI mTLS, NF integrity verification
Quantum Subscriber Data	Harvest-now decrypt-later on HLR/HSS, Ki keys, RAND/SRES vectors	Billions of subscriber records exposed, GDPR/NIS2 fines up to 4% revenue	ML-KEM/ML-DSA at network layer, no device update needed, crypto-agile

Threat 1: SS7/Diameter Legacy Protocol Exposure

SS7 (Signaling System No. 7), including MAP, ISUP, and TCAP protocols, underpins virtually all mobile voice and SMS services globally. Despite being designed in 1975 with no authentication, it remains the backbone of inter-carrier communication. Diameter, its 4G-era successor, improved some aspects but inherited fundamental trust model weaknesses.

- **Location tracking** of any mobile subscriber is available for purchase at \$500 per target
- **SMS interception** enables bypass of SMS-based MFA across banking, government, and enterprise applications
- **Call diversion** allows silent call forwarding to adversary-controlled numbers
- **850M+ attacks recorded** in 2024 — averaging nearly 100 SS7 attacks per second globally
- **Subscriber denial of service** attacks can silence journalists, activists, and government officials

Threat 2: 5G Hyperscale Attack Surface

5G networks introduce unprecedented complexity. The 5G core Service-Based Architecture (SBA) relies on HTTP/2 RESTful APIs between network functions — AMF, SMF, UPF, AUSF, UDM, PCF — connected via the N2, N3, N4, N6, and SBI interfaces. Each interface is a potential attack vector. Each network slice has different security requirements that a one-size-fits-all approach cannot address.

- A compromised 5G core network function can expose subscriber session data for **millions of users simultaneously**
- Cross-slice attacks allow lateral movement from low-security IoT slice to high-security emergency-services slice

- PFCP manipulation on N4 interface can redirect user-plane traffic at carrier scale
- 5G core has **3x more attack surface** than 4G, with larger proportion internet-adjacent

Threat 3: Quantum Threat to Subscriber Databases

Mobile network operators maintain some of the most sensitive databases on earth: subscriber identity (IMSI/MSISDN), authentication credentials (Ki keys, RAND/SRES vectors), location history, communication metadata, biometric data for identity verification, and financial information for billing. This data is protected today by classical cryptography — AES-128 for 4G subscriber authentication, RSA and ECDH for key exchange — all of which quantum computers will break.

- Nation-state adversaries harvesting encrypted subscriber data today for **quantum-era decryption later**
- Authentication credentials from quantum-decrypted HLR/HSS enable subscriber impersonation at scale
- GDPR/NIS2 penalties for subscriber data breach can reach **4% of global annual turnover**
- GSMA FS.19 has flagged quantum vulnerability as a critical long-term risk requiring immediate roadmap

QBITEL BRIDGE — SEVEN CORE CAPABILITIES

Full-spectrum protection across every telecom protocol and domain

01

SS7/Diameter Protocol Security

Real-time MAP filtering blocks location tracking, SMS interception, and call diversion attacks in under 1 second. PQC-wrapped authentication vectors protect HLR/HSS subscriber data. Covers E1/T1 legacy and SIGTRAN/M3UA IP-encapsulated SS7. Immutable blockchain audit trails for GSMA FS.11/FS.19 compliance.

02

5G Core Network Slice Protection

Per-slice PQC policy enforcement for eMBB, URLLC, and mMTC slices with independent key hierarchies. mTLS with PQC hybrid key exchange on all SBI interfaces. AMF/SMF/UPF/AUSF/UDM/NRF integrity verification. NSSAI manipulation detection and slice admission control.

03

SIP/VoIP Fraud Prevention

Real-time IRSF detection with sub-second blocking of calls to international premium rate numbers. Wangiri one-ring fraud detection. SIP INVITE flood rate limiting for 100,000+ concurrent calls. ML-DSA-signed SIP trunk authentication. Compatible with Cisco CUBE, Ribbon SBC, Oracle ACME Packet.

04

IoT/mMTC Mass Device Security

Network-layer PQC protection for 50M+ IoT devices — no firmware update required. ML-KEM optimized for NB-IoT, LTE-M, and eMTC constrained devices. Real-time IoT botnet detection with automatic quarantine. GSMA IoT security guidelines and ETSI EN 303 645 alignment.

05

Carrier-Grade PQC at Scale

150,000+ PQC operations per second per node with hardware acceleration. ML-KEM-768/1024, ML-DSA-65/87, SLH-DSA-SHAKE-256 support. FIPS 140-3 Level 3 HSM integration. Linear horizontal scaling. Sub-millisecond cryptographic latency invisible to subscriber experience.

06

Autonomous Fraud Detection

Graph neural network models trained on inter-carrier signaling graphs detect coordinated attacks across multiple operators. 500+ behavioral features per subscriber. Federated learning preserves data sovereignty. False positive rate below 0.01% — carrier NOC ready. Covers IRSF, SIM swap, bypass fraud, wangiri, roaming fraud, and PBX hacking.

07

Regulatory Compliance Automation

Continuous automated generation of 3GPP TS 33.501, GSMA FS.19/FS.11, NESAS, NIS2, FCC/CISA, ETSI NFV-SEC, BEREC, and GDPR compliance evidence. Daily cryptographic logs with HSM attestation. NIS2 72-hour incident reporting. Monthly SS7 attack blocking reports in GSMA format.

COMPLIANCE COVERAGE MATRIX

Automated evidence generation for every major telecom regulatory framework

Regulation	Scope	Key Requirements	QBITEL Bridge Status
3GPP TS 33.501	5G security	SBI mTLS, SUPI protection, slice security	Full coverage
GSMA FS.19	PQC readiness	Algorithm migration, subscriber protection	Full coverage
GSMA FS.11	SS7 security	MAP filtering, location privacy	Full coverage
NESAS / SCAS	Equipment assurance	Security test evidence	Automated evidence
NIS2 Directive	EU critical infra	Incident reporting, security measures	Automated reporting
FCC SS7 Action	US carriers	SS7 monitoring, remediation	Full coverage
ETSI NFV-SEC	Virtual NFs	vNF integrity, isolation	Full coverage
BEREC Security	EU telecom	Availability, integrity measures	KPI automation
GDPR Article 32	EU data protection	Encryption of subscriber data	PQC encryption

INTEGRATION ECOSYSTEM

Nokia, Ericsson, Open RAN, Amdocs, Oracle Communications — and more

Vendor / Platform	Integration Points	Deployment Mode
Nokia	CloudBand, AVP telemetry, NetAct/1Network Manager, Bell Labs PQC research	Cloud-native, containerized
Ericsson	Cloud Manager, ERIC-OSS, Radio System O-RAN xApp, UDM/AUSF PQC	Cloud-native, hybrid
Open RAN / Free5GC	O-RAN O1/A1/E2, Free5GC, Open5GS, OpenShift Telco, ONAP	Kubernetes-native
Amdocs	CES fraud alerts, Revenue Management, Network Cloud OSS northbound API	API integration
Oracle Communications	Diameter Signaling Router, Session Border Controller, PCRF/PCF	Inline / API
Subex / TEOCO	Fraud Management System REST API, real-time event feed, revenue impact	API integration
Syniverse	Roaming hub integrity, inter-carrier settlement, fraud data sharing	Federated

PERFORMANCE SPECIFICATIONS

Carrier-grade SLAs engineered for the world's most demanding networks

Metric	Specification	Notes
PQC Operations/Second	150,000+	Per node, hardware-accelerated
SS7 Attack Block Latency	<1 second	From detection to blocking action
SIP Call Processing	100,000+ concurrent	No measurable latency addition
5G Slice Policies	Unlimited	Per-slice, per-subscriber granularity
Subscriber Capacity	500M+	With horizontal node scaling
Availability SLA	99.999%	Five-nines, active-active clustering
Cryptographic Latency	<1ms	Sub-millisecond, invisible to subscriber
Fraud Detection Latency	<500ms	Real-time blocking capability
False Positive Rate	<0.01%	AI-tuned, carrier NOC ready
HSM Key Operations/sec	10,000+	Hardware-bound, FIPS 140-3 L3

COMPETITIVE DIFFERENTIATION

Why QBITEL Bridge — not point solutions

The telecom security market is fragmented across specialist vendors: signaling firewall vendors (ISMS, Cellusys, Mobileum), fraud management systems (Subex, TEOCO, Syniverse), network security vendors (Palo Alto, Fortinet), and emerging PQC vendors. Each solves part of the problem. QBITEL Bridge is the only platform that addresses all dimensions simultaneously.

DIFFERENTIATOR	CAPABILITY
Unified Platform	Protocol security + PQC + fraud detection in one integrated platform — eliminating integration complexity, data silos, and coverage gaps of point-solution architectures
True Carrier-Grade Performance	150,000+ ops/sec PQC with 99.999% availability — not lab-benchmark figures that fall apart under real traffic
No Device Updates Required	Network-layer PQC protection means 500M+ legacy devices are protected without a single firmware update or SIM swap
Full Protocol Stack Coverage	From 1975-era SS7 through Diameter, SIP, GTP, PFCP, and 5G SBI — single policy framework and unified audit trail

Compliance Automation	3GPP, GSMA, NIS2, and FCC-ready documentation generated automatically — eliminating hundreds of audit hours per cycle
Any-Environment Deployment	On-premises, private cloud, AWS/Azure/GCP, and hybrid — with Kubernetes-native telco cloud containers

CUSTOMER SCENARIOS

Real outcomes for Tier-1 MNOs, MVNOs, and telecom equipment vendors

Scenario 1: Tier-1 MNO — Quantum-Safe Subscriber Database Protection

Organization	Challenge	Solution	Outcome
European Tier-1 MNO, 80M subscribers, 12 countries	Nation-state adversary harvesting SS7 signaling data for quantum-era decryption. NIS2 compliance deadline. GSMA FS.19 quantum readiness assessment due.	PQC wrapping of SS7 MAP auth vectors. Retrospective analysis found 127 undetected VIP surveillance campaigns. 5G SA core with ML-DSA SBI. Automated FS.19 documentation.	GSMA FS.19 certified in 14 weeks. All SS7 surveillance blocked. NIS2 evidence provided to regulator. EUR 40M in projected fines avoided.

Scenario 2: MVNO — IRSF Revenue Protection

Organization	Challenge	Solution	Outcome
UK MVNO, 4M subscribers, operating over Tier-1 host MNO	EUR 2.3M/month IRSF losses. Fraud tools only flagging after call completion — too late for real-time blocking. Contractual liability to host MNO for fraud traffic costs.	SIP trunk IRSF detection engine. ML models trained on 6 months of CDR history. Sub-second blocking of IRSF calls. Wangiri detection. Automatic trunk suspension.	IRSF reduced from EUR 2.3M/month to EUR 15K/month — 99.3% reduction in 8 weeks. ROI in 60 days.

Scenario 3: Equipment Vendor — PQC-Ready Product Portfolio

Organization	Challenge	Solution	Outcome
Tier-2 SBC and signaling gateway vendor, global MNO customer base	MNO RFPs requiring PQC capability. Engineering team estimated 18 months for native PQC. Risk of losing major contracts worth tens of millions of euros.	QBITEL Bridge PQC SDK integrated into SBC firmware as OEM module. Pre-integrated PQC key management. Co-branded joint MNO sales enablement.	Three major MNO contracts won citing PQC differentiation — EUR 47M total value. Time-to-market reduced from 18 months to 12 weeks.

DEPLOYMENT TIMELINE

From discovery to full carrier-grade protection

Week	Phase	Key Deliverable
1-2	Discovery & Assessment	Network topology map, protocol inventory, SS7/Diameter threat profile
3-4	Architecture Design	Integration architecture, PQC policy framework, HSM design
5-6	Infrastructure Provisioning	HSM installation, network taps, pilot environment setup
7-9	Protocol Security Configuration	SS7 MAP filtering rules, Diameter policies, SIP trunk hardening
10-11	5G Core Integration	SBI mTLS/PQC, slice policies, AMF/SMF/UPF/NRF protection activation
12-13	Fraud Detection Activation	ML model initialization, IRSF rules, FMS integration, wangiri
14-15	Monitoring & Alerting	NOC dashboards, GSMA KPI reporting, NIS2 regulatory feeds
16	Go-Live & Acceptance	Full production cutover, SLA validation, runbook handover

MVNO Accelerated Deployment: 6 weeks to full SIP fraud protection and compliance reporting, using pre-integrated connectors for major host MNO platforms.

NEXT STEPS

Start your quantum-safe telecom journey today

STEP	ACTIVITY	DESCRIPTION
Step 1	Threat Exposure Assessment (Complimentary)	Passive analysis of SS7 and Diameter signaling traffic — identifying active attack campaigns, protocol anomalies, and quantum vulnerability exposure with no network changes. Confidential executive briefing within 5 business days.
Step 2	PQC Readiness Assessment	Structured assessment of your cryptographic posture across all network domains — mapping current algorithm usage, identifying quantum-vulnerable systems, and producing a prioritized migration roadmap aligned with GSMA FS.19 requirements.
Step 3	30-Day Pilot Program	Live pilot on a defined network segment — signaling link, 5G core slice, or SIP trunk group — demonstrating real attack detection, fraud prevention impact, and compliance evidence generation with your actual traffic.

Step 4	Full Network Deployment	Full deployment with QBITEL Bridge-certified implementation support, 24/7 NOC monitoring, ongoing threat intelligence feeds, and continuous compliance automation.
---------------	--------------------------------	--

Enterprise Sales enterprise@qbitel.com	Technical Pre-Sales https://bridge.qbitel.com	Partner Program OEM, VAR & SI inquiries welcome
--	---	--

Securing the networks that secure the world.